# Does latent training improve token-level chemical meaning?

**Token-aware chemical readouts favor the parent.**
The latest checkpoint is easier to reconstruct sparsely and more resistant to peak removal, but neither establishes better chemical alignment.

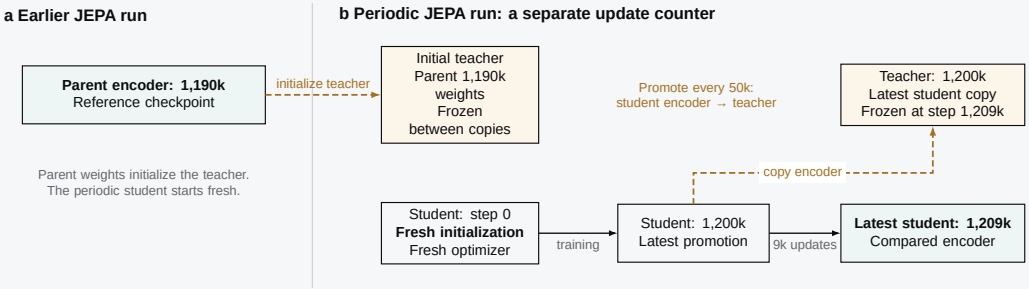
0.7987 → 0.7750

Final-token MACCS AUROC: parent 1,190k / periodic student 1,209k

> Attention helps both encoders; sparse autoencoders and causal tests refine what the difference means.

Native JAX checkpoint comparison. 1,024 held-out molecules; ensemble of three token-attention readouts. Parent initializes the periodic teacher, not the student.

# These checkpoints come from different training histories




Green boxes are compared. Different run counters: **not 19k additional updates on the parent weights**.

Solid: student training. Dashed: weight copy. Teacher receives no gradients; promotions keep student + optimizer state. k = 1,000 updates.

> The periodic student starts fresh; this is not 19,000 further updates of the parent encoder.

Pinned latest committed checkpoint: 1,209,000. Parent: 1,190,000; stored periodic teacher: 1,200,000. Teacher copies every 50,000 updates. No matched nonpromotion control was run.

# Separate feature discovery from held-out evaluation

## Train

**4,096**

Fit readouts
Train SAEs
Fit normalization
Align replicas

## Validation

**1,024**

Tune readouts
Freeze features
Select layers
Set thresholds

## Test

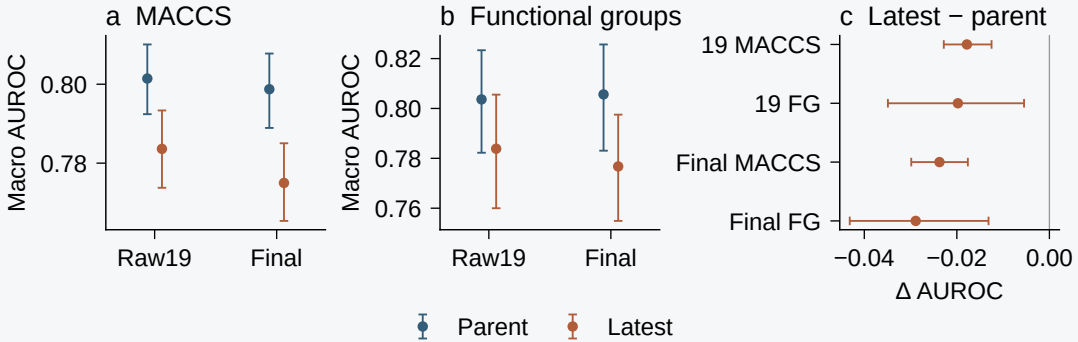
**1,024**

Score chemistry
Run interventions
Check controls
Keep failures

One representative spectrum per connectivity identity; all candidate choices precede test evaluation.

MassSpecGym 1.5: 4,096 / 1,024 / 1,024 train / validation / test molecular connectivities. Frozen native JAX encoders. 166 MACCS bits, seven elements, 15 SMARTS functional groups. Highest-activation example cards are separately labeled post-confirmation illustrations.

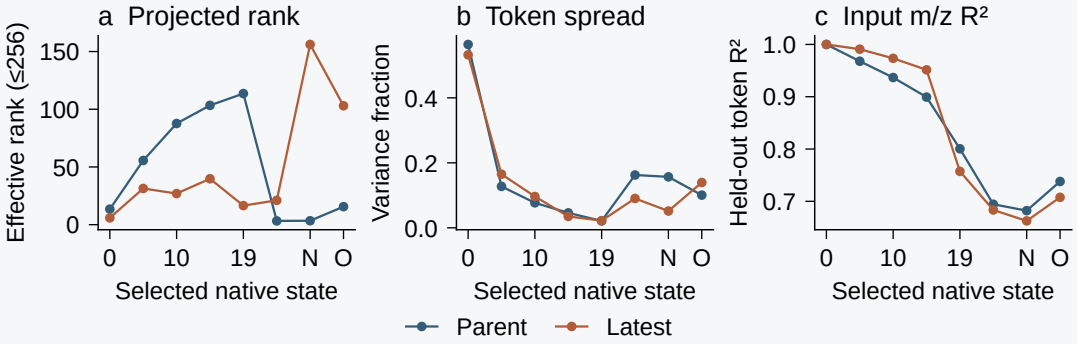
# Learned token attention does not reverse the parent advantage




**Latest block-19 attention gains 0.0052 MACCS AUROC over mean pooling; the parent remains stronger.**

Physical peak tokens; CLS excluded. Three readout seeds; learning rate and epoch selected on validation. AUROC of mean seed logits; 1,024 test molecules. 145 MACCS and 13 functional-group targets with shared support. Paired molecule bootstrap, descriptive 95% intervals.

# Representation shape changes with depth and normalization

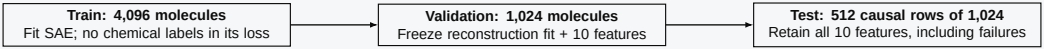



A low-rank layer and a high-rank final output can coexist; rank alone does not measure chemical meaning.

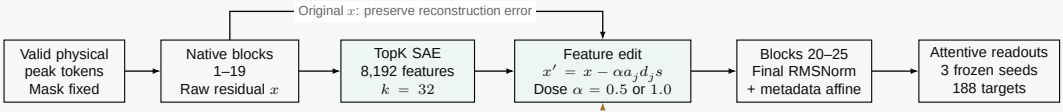
Geometry at eight selected states across all 6,144 panel molecules; rank uses a fixed 256-dimensional projection. Held-out input decoding. N: final RMSNorm; O: final affine output. Pooled chemical probes cover all 28 states.

# Decompose token activations, then intervene in the native encoder

**a Molecule-disjoint fitting and frozen feature selection**




**b Residual-preserving feature edits in the native encoder**

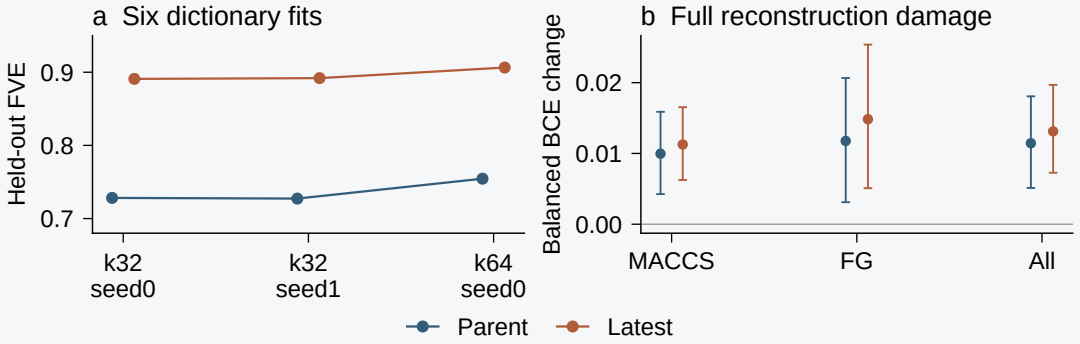



Solid: computation / selection. Dashed: control edit. Valid peaks only; CLS + padding unchanged. Encoder and decoder weights stay fixed.

**Feature removal preserves the reconstruction residual; controls use the same requested token amplitudes.**

Raw block 19; 8,192 SAE features; train-only normalization; equal molecule weight. No chemistry labels enter SAE training. Half/full feature removal; firing-matched feature and orthogonal random directions. Three fixed final-token readouts. No positive steering or sufficiency test.

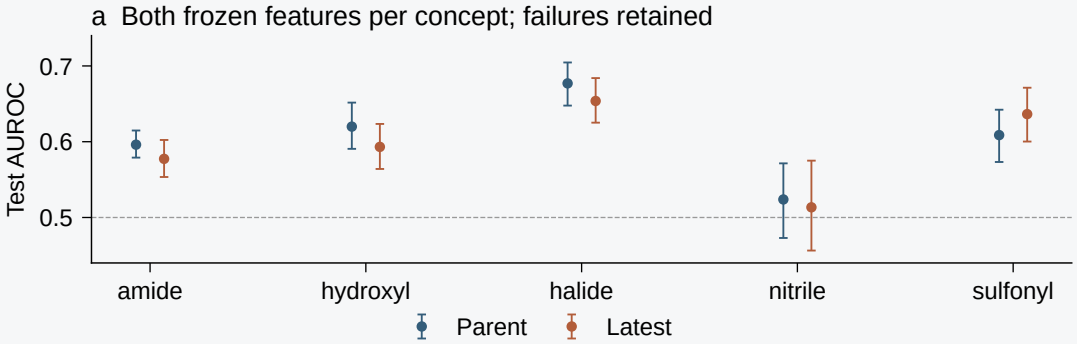
# The latest token states are more sparsely reconstructable




At 32 active features per token, held-out variance explained rises from 72.8% to 89.1%.

Raw block 19; 8,192 SAE features; train-only normalization; equal molecule weight. No chemistry labels enter SAE training. Two seeds at k=32 and one k=64 control per encoder. 22,474 sampled test tokens with equal molecule weights. All selected fits reached the 40-epoch budget; convergence is not established.

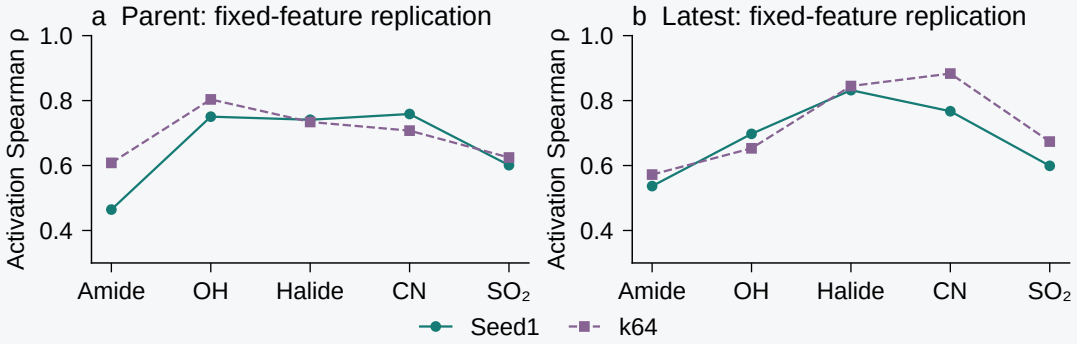
# Useful halide features coexist with weak chemical candidates




Ten candidates per encoder were frozen before test; the failed nitrile features remain in the analysis.

Two candidates each: amide, hydroxyl, halide, nitrile, sulfonyl. No amine candidate passed. Test AUROC uses molecular maximum feature activation; dense-channel and random-dictionary baselines are available.

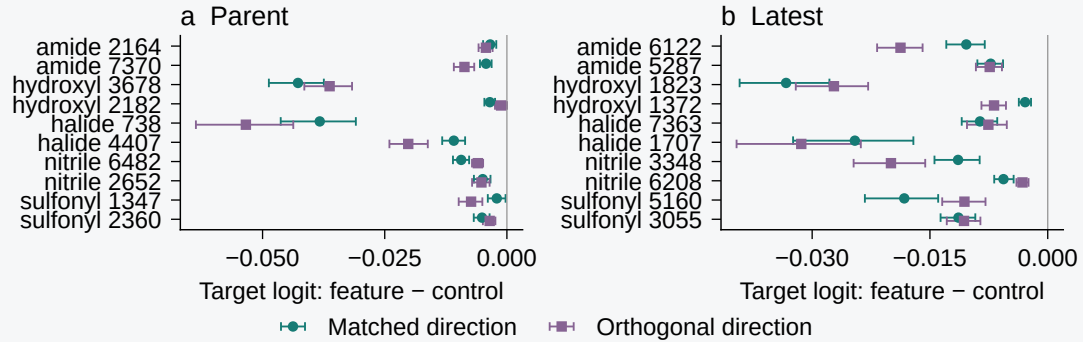
# Stability is necessary evidence, but it does not guarantee chemistry




Halide features replicate; stable activations can also encode acquisition or peak-count variation.

Train-only, label-free feature alignment; test activation Spearman. Each point averages two frozen features per concept. IDs are dictionary-local; two latest hydroxyl features share one replica.

# Selected sparse directions influence the chemical readouts




38 of 40 full-dose feature-versus-control contrasts pass Holm correction; chemical specificity is weaker.

512 independent test molecules. Mean logit shifts across three fixed readouts. 20 frozen features × two controls; paired t approximations with Holm correction. Bootstrap intervals are descriptive.

# Causal readout influence can coexist with failed recognition

## Latest nitrile feature f3348

Held-out chemical association
**AUROC 0.496**

Full feature removal
**Nitrile-logit change** $-0.0348$

## Reconstruction damage

Full SAE reconstruction:
Balanced BCE increase
**Parent** $+0.01145$
**Latest** $+0.01312$

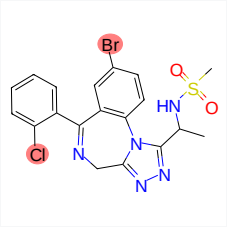
Other labels and negative molecules also respond.

A feature can drive a chemical prediction without reliably detecting that chemical group.

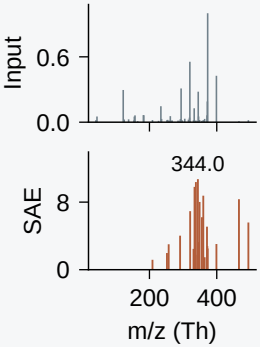
Association: 1,024 test molecules. Causal/reconstruction effects: 512 molecules; train- and test-supported targets. Latest f3348 is a prespecified failed candidate, retained without reselection.

# A replicated halide feature still has strong false positives

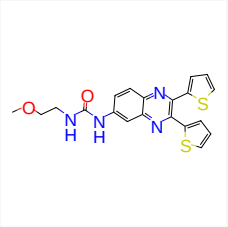
a Concept present



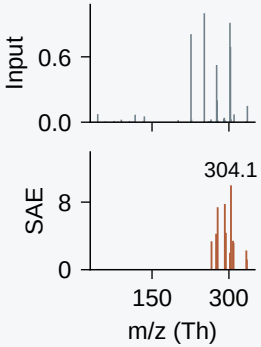

CE 45
53 peaks




b False positive



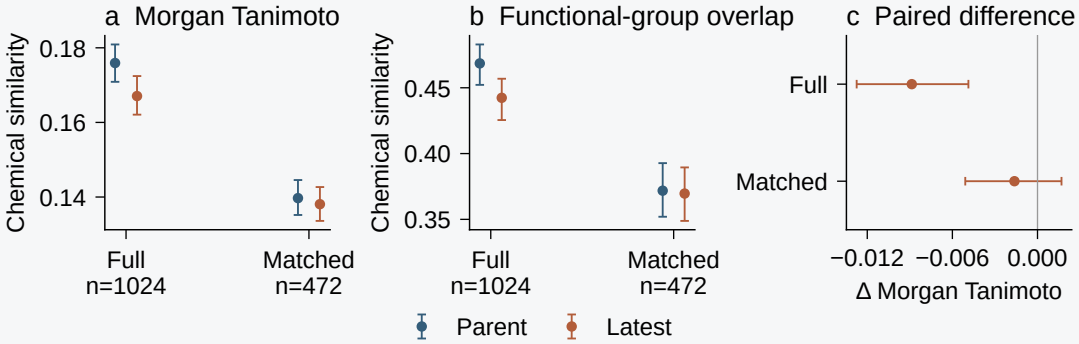

CE 60
43 peaks




Chemical motifs label the molecule; activated peaks are not validated assignments to those motifs.

Latest raw-19 SAE feature f7363. Positive and false-positive examples follow the frozen atlas selection: highest test activation within each class, chosen after confirmation. Measured spectra; illustrative cases.

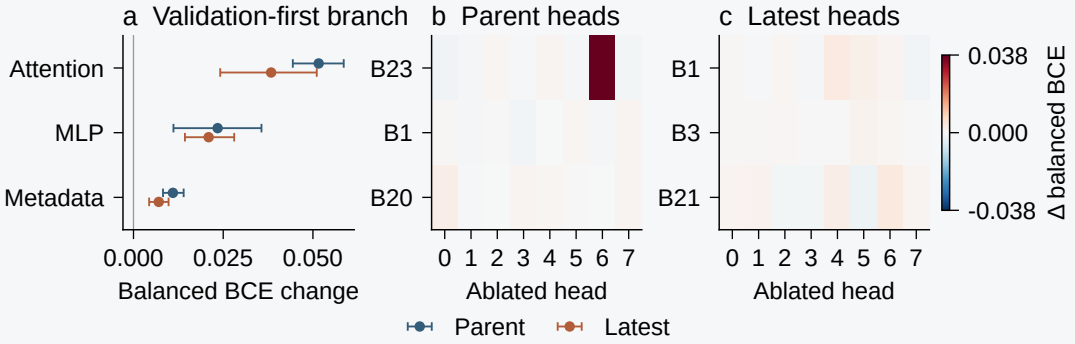
# The full-bank decline weakens after nuisance matching




**Within strict mass and acquisition controls, the checkpoint difference is inconclusive.**

Test-to-train SAE-max cosine neighbors. Full: 1,024 queries; matched: 472 with at least two candidates. Same instrument/adduct/CE presence; 5 Da / 10 CE / 5-peak calipers. Δ: latest minus parent.

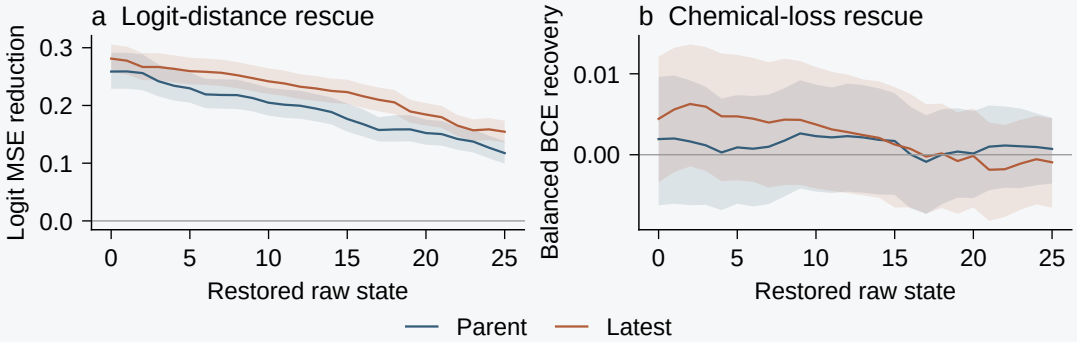
# Branch effects and individual-head effects differ




The parent's block-23 head 6 has a strong effect; head contributions are not an additive decomposition.

All 25 attention/MLP/conditioning branches screened on 256 validation molecules. Three layers per branch frozen, then tested on 512 molecules; 24 heads per encoder. Test rankings are descriptive.

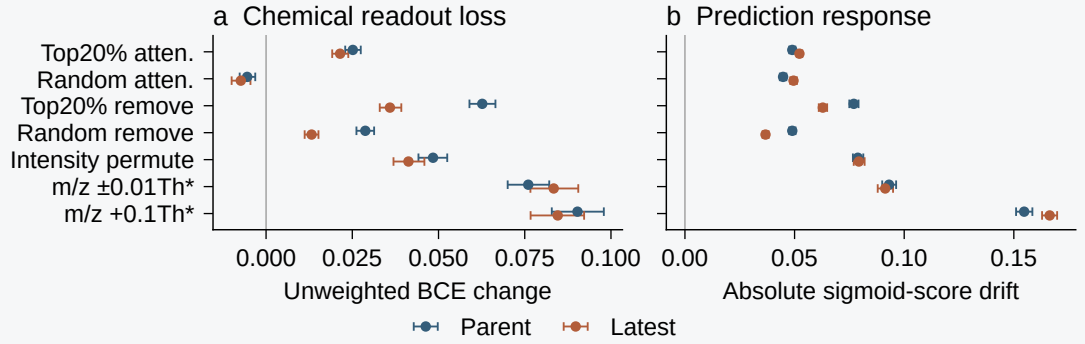
# Restoring corrupted peak states recovers the original output




**Selected-token rescue beats random slots, but chemistry-loss recovery is inconclusive in this cohort.**

256 test molecules; top-20% peak intensities multiplied by 01. Curves show selected-minus-random recovery from clean-state patches at raw depths 0–25; equal slot counts. Full-state rescue is exactly clean.

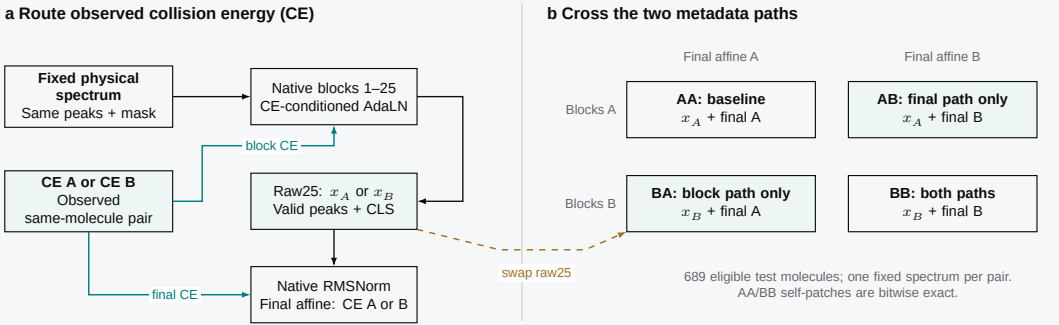
# Peak-removal damage decreases, while clean error remains higher




Smaller corruption damage does not reverse the parent advantage in absolute chemical prediction.

1,024 test molecules; ordinary BCE averaged over 177 train-active outputs using mean sigmoid scores across readouts. Different from class-balanced causal BCE. Paired molecule intervals.
* Artificial $m/z$ perturbations.

# Route observed collision energy through two separate paths

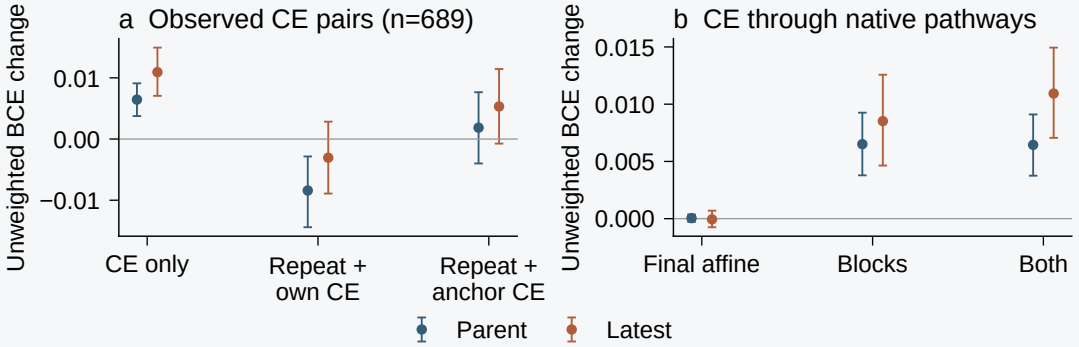



Black: computation. Teal: CE input. Dashed: activation replacement. No molecule / spectrum edit; the same three frozen readouts score all four cells.

**The same measured spectrum receives old or observed-repeat energy through block and final conditioning.**

689 eligible same-molecule repeats. Raw block-25 patching separates upstream conditioning from the final affine path. All valid slots, including CLS, are patched. Both diagonal self-patches are exact.

# The latest chemical readouts are more sensitive to collision energy




**Near-zero average final-path loss change does not imply zero response in individual predictions.**

Observed CE swap on 689 shared molecules. Ordinary BCE over 177 train-active outputs; paired molecule intervals. Path interaction includes nonlinear encoder/readout/loss effects. Repeat cosine is readout-logit cosine.

# Compressibility, chemistry and robustness give different answers

| Question | Latest versus parent |
|---|---|
| Token-attention chemistry | Parent stronger |
| Sparse reconstruction | Latest stronger |
| Matched chemical neighbors | Difference inconclusive |
| Peak-removal robustness | Latest loses less, but starts worse |
| Collision-energy invariance | Latest more sensitive |

> The evidence does not support an overall chemical-semantic improvement at this endpoint.

Checkpoint comparison conditional on the measured panel, readouts and interventions. These results do not isolate the causal effect of the training recipe or promotion schedule.

# The interventions reproduce the actual cached encoder function

**Primary computation**
Native JAX, BF16 residuals, H100, batch 8. Full encoder trace; readouts compiled separately.

**Exact checks**
Both caches match native pilots bitwise. Identity patches and inactive-feature edits change no logits.

**Why execution details matter**
Latest native batch 16 differs from batch 8 by 2.585% relative L2. PyTorch execution also differs despite identical weights.

**Feature checks**
19,390 physical-token codes per arm match exports exactly; realized BF16 control norms are only approximately matched.

Archived batch-16 and older PyTorch results are excluded from this report's primary measurements.

Native pilot, extraction, precision diagnostic, causal manifests and sparse-code audit. No claim of exact TPU–H100 equivalence. Full requested feature/control norms match before BF16 casting.

# A tested mechanism is narrower than a complete chemical circuit

**Established**
Token-level chemical signal.
Replicated sparse associations.
Feature and component effects.
Token rescue and metadata paths.

**Not demonstrated**
Positive steering or sufficiency.
Head-to-feature mediation.
Validated peak/fragment links.
An exhaustive chemical circuit.

**Unresolved**
Training-recipe causality.
Pretraining overlap.
Rare chemistry and generalization.

**Next tests**
Matched training controls.
Independent chemistry data.
Fragment labels and path mediation.

Keep the parent as the chemistry reference and retain token-aware evaluation before claiming a gain.

One checkpoint per training history. Molecule-bootstrap uncertainty is conditional on fixed models; most secondary intervals are unadjusted. Highest-activation examples are illustrative, not new confirmation.

# Figures and conclusions trace back to the completed study

**Study artifacts**

`artifacts/encoder_chemistry_20260930/`

Checkpoints, readouts, six SAEs, interventions, cases and independent audit.

**Editable report and rebuild**

`presentations/encoder_chemistry/encoder_chemistry.tex`

`bash presentations/encoder_chemistry/build.sh`

Vector figures, compact inputs and source bundle included.

**Methods:** Gao et al. (2024), SAEs; Zhang & Nanda (2023), activation patching. Chanin et al. (2024), feature splitting and absorption.

Every measured chart uses saved results; explanatory diagrams are labeled as computation or intervention.

arXiv:2406.04093; arXiv:2309.16042; arXiv:2409.14507. Report date: 30 September 2026.